Escuela de Ingeniería Informática
Pontificia Universidad Católica de Valparaíso
Avenida Brasil 2950, Valparaíso, Chile
ignacio.araya@pucv.cl

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Editors
Computers & Operations Research

Dear Editors,

Please find enclosed the manuscript *"A beam search for the air cargo palletization problem with weight distribution constraints"*, which I submit for consideration for publication in Computers & Operations Research, in the area *Optimization – Approximate Methods*.

The manuscript addresses the palletization of air cargo into unit load devices (ULDs), a three-dimensional multiple bin size bin packing problem with non-cuboid bins, a floor edge, weight capacities, fragile boxes, support constraints and center of gravity constraints, derived from real data of a Boeing 777. It continues a line of work on block-based beam search for container loading published in Computers & Operations Research (Araya and Riff, 2014; Araya, Guerrero and Nuñez, 2017).

The paper is not a direct application of that method: a container loading algorithm does not handle the features of this problem, and the contribution lies in the components that make the approach work for air cargo:

- blocks that are built and placed so that the support and fragility constraints hold, with the support criteria and the padding material of the problem;
- the handling of non-cuboid ULDs, by shifting blocks along the bevelled contours, and of the floor edge and the substructure, as fixed obstacles with dormant free spaces;
- the center of gravity inside the search, through a penalty in the value of the solutions, combined with a translation of the load within an interval of feasible translations;
- a multi-ULD framework whose reduction phase replaces each ULD by at most two smaller ones.

The algorithm is evaluated on the 300 benchmark instances in the three settings of the current state of the art (Heßler et al., European Journal of Operational Research, 2025), with parameters tuned once on a separate training set, an instance-by-instance comparison at the same hardware-normalized running time, and an ablation study. It improves on the state of the art in both volume and weight distribution: at a comparable running time it uses 3–4% less ULD volume, and 7–8% less with a few seconds per instance, while at most 1% of its ULDs violate the center of gravity constraint, against 7–11% for the previous method. The experiments also show that most of the savings come from the reduction phase rather than from the construction.

All the solutions are checked by an independent validator, and the code, the instances, the solutions and the scripts that reproduce the tables and figures are publicly available at https://github.com/rilianx/bsg-uld.

This manuscript is original, has not been published, and is not under consideration for publication elsewhere. The author declares no competing interests.

Sincerely,

Ignacio Araya
Escuela de Ingeniería Informática
Pontificia Universidad Católica de
Valparaíso